

Drone Phenology Monitoring

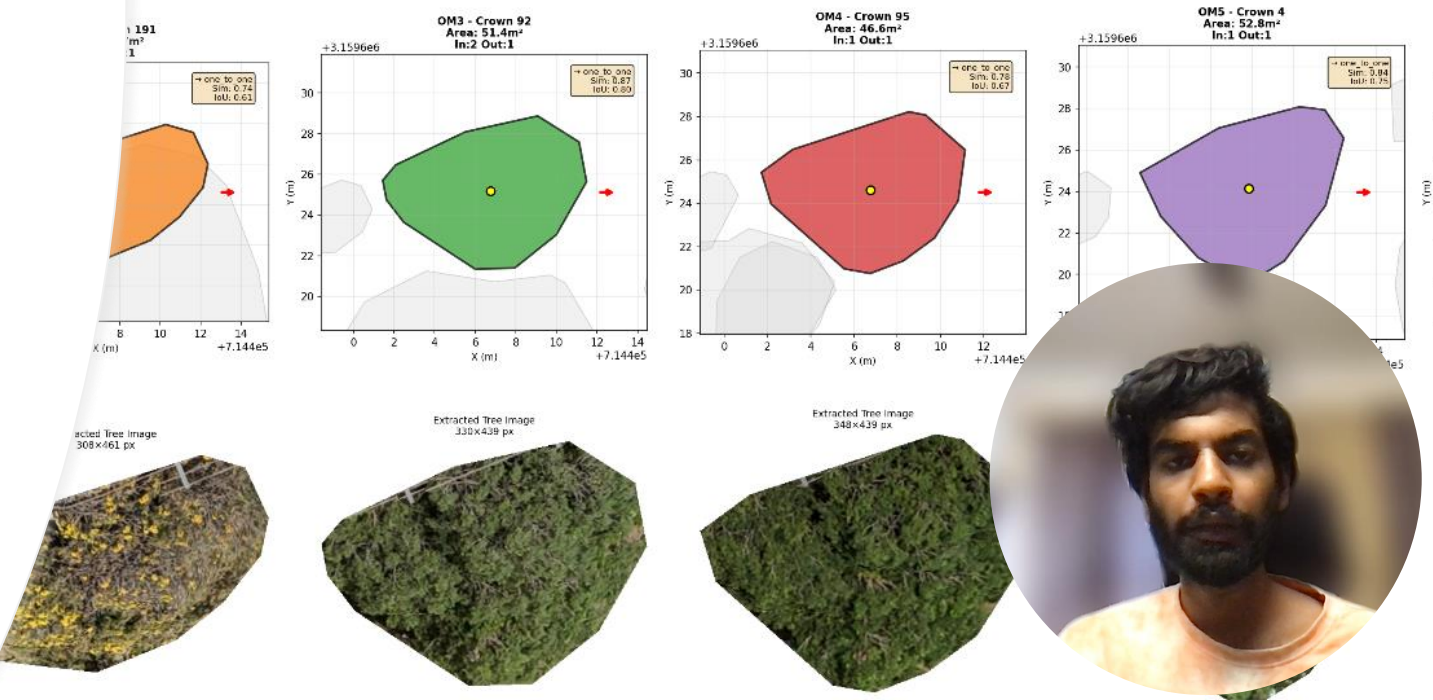
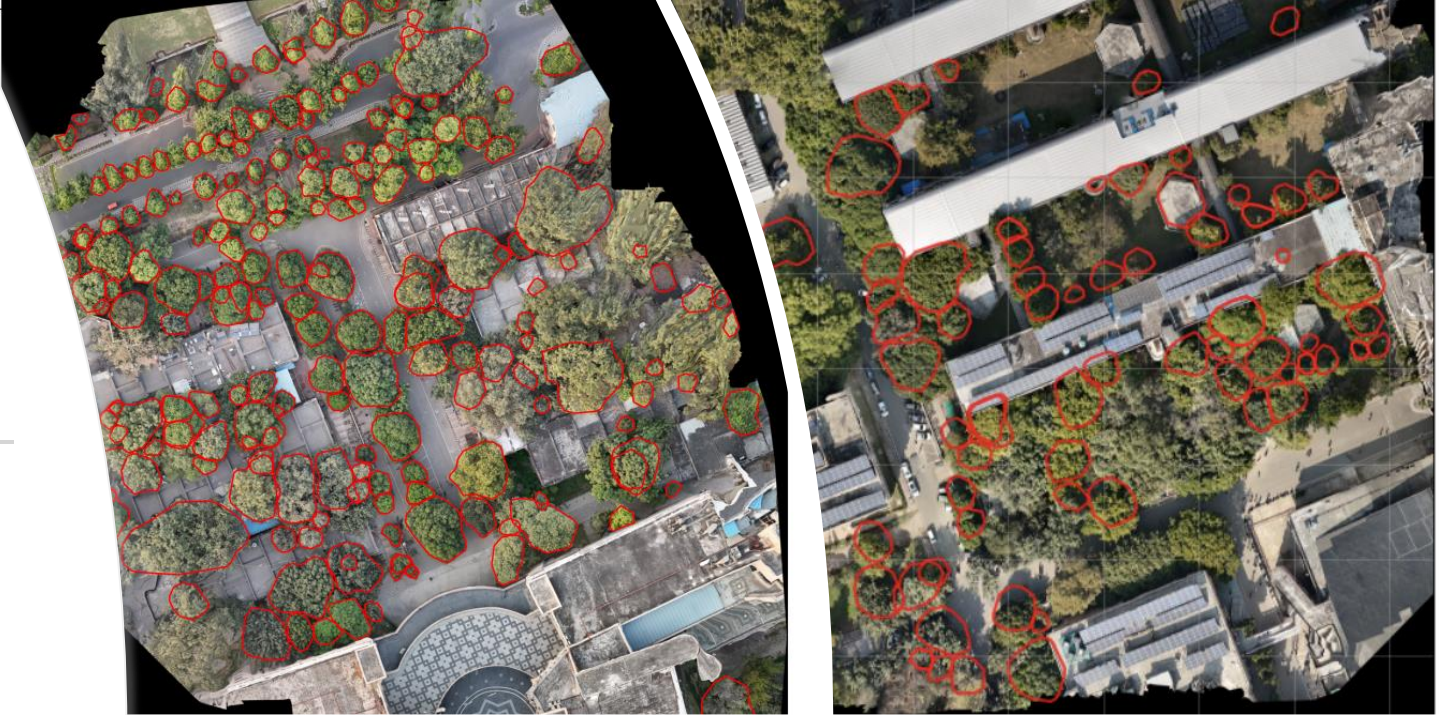


Prof. Aaditeshwar Seth
Parth Thakur, Gaurav Rajput
2021CS0615, 2024CSY7558

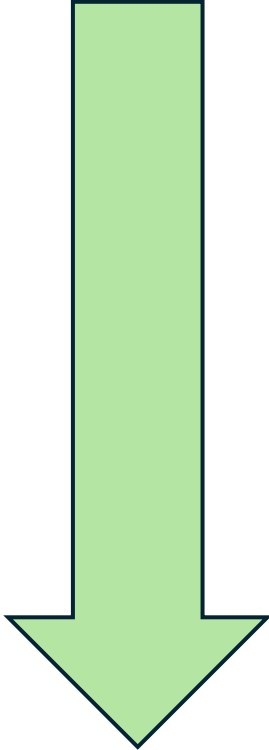


Overview

- Goal: Develop an automated system for monitoring tree cover using drones
- Capture Aerial Images
- Generate Orthomosaics
- Tree Crown Detection
- Temporal tracking of Individual Trees
- Phenological Monitoring



Work Done Previously



Planned flight paths and executed repeated drone surveys over LHC and SIT areas

Drone shots processed in WebODM to produce a georeferenced orthomosaic(stitching)

Tree crown polygons detected over the orthomosaic using Detectree2

Graph-Based Tracking Framework

- Nodes: Individual crown detections (one node per tree per date)
- Edges: Connect matching crowns across consecutive orthomosaics
- Matching criteria: Spatial proximity, geometric overlap (IoU), shape similarity

Conditional Matching based on overlap cases & Chain Resolution



Multi-Threshold Crown Detection

- Tree crown detection confidence varies across orthomosaics due to illumination changes, seasonal phenology, and image quality
- Single threshold approach misses detections or creates false positives
- Captures trees that appear/disappear at different confidence levels
- Better handles seasonal changes in canopy appearance
- We maintain a multi-threshold crown cache; Query lower confidence layers when high-confidence matching fails

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{
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  "gpkg_path": "/Users/hbot07/VS Code/Drone-Phen",
  "thresholds": [
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    0.55,
    0.6,
    0.65
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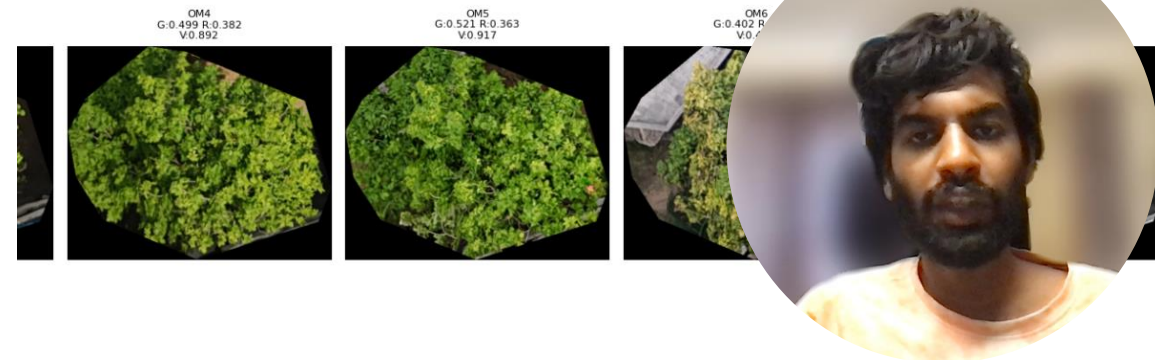


Phase Cross Correlation Alignment

- Orthomosaics exhibit spatial misalignment due to GPS drift, wind, etc; same physical tree appears at different coordinates across dates
- With OM1 fixed as spatial anchor, we do FFT-based subpixel alignment on grayscale rasters (downsampled for speed)
- Calculate cross-power spectrum; Inverse FFT → correlation surface with peak at $(\Delta x, \Delta y)$; Apply affine translation
- IoU recovery: +0.15 mean IoU increase for same crown pairs
- Match rate improvement: +18% (from 78% → 96%)

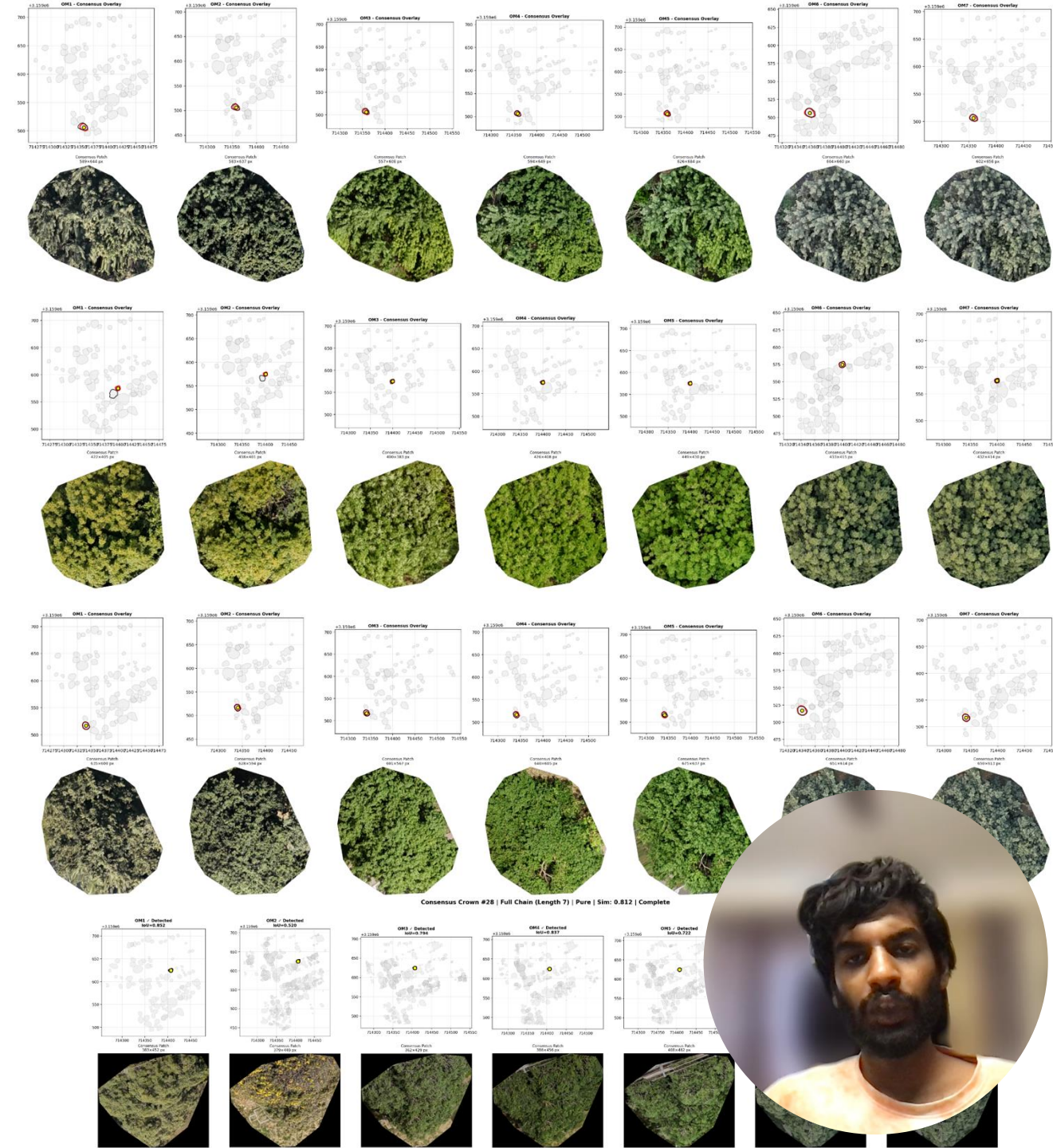
OM	Δx (m)	Δy (m)	Distance (m)
1	0.0	0.0	0.0 (ref)
2	+12.3	-8.7	15.1
3	+18.9	-15.2	24.3
4	+31.4	-22.1	38.4
5	+28.6	-19.8	34.5
6	+35.2	-24.9	43.1
7	+41.7	-29.3	51.0

Tree 12 - Consensus Crown Across OMs



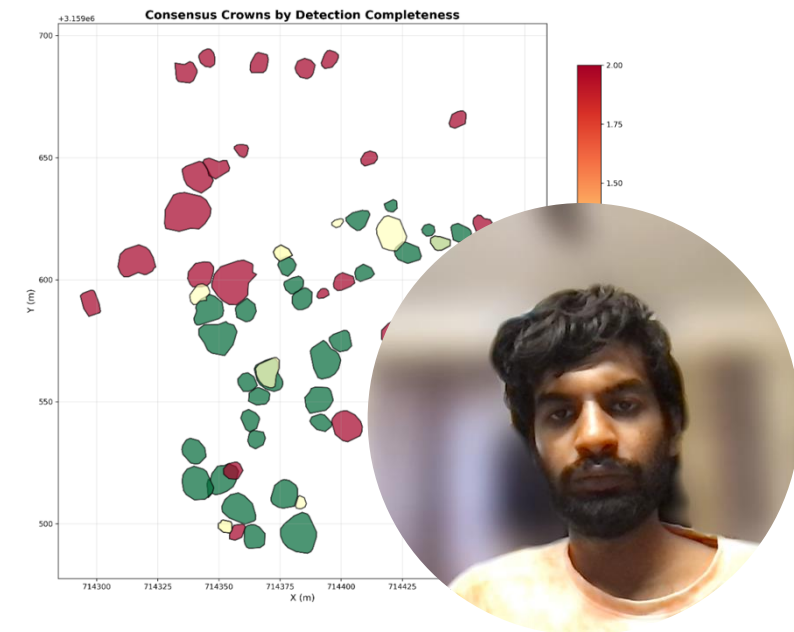
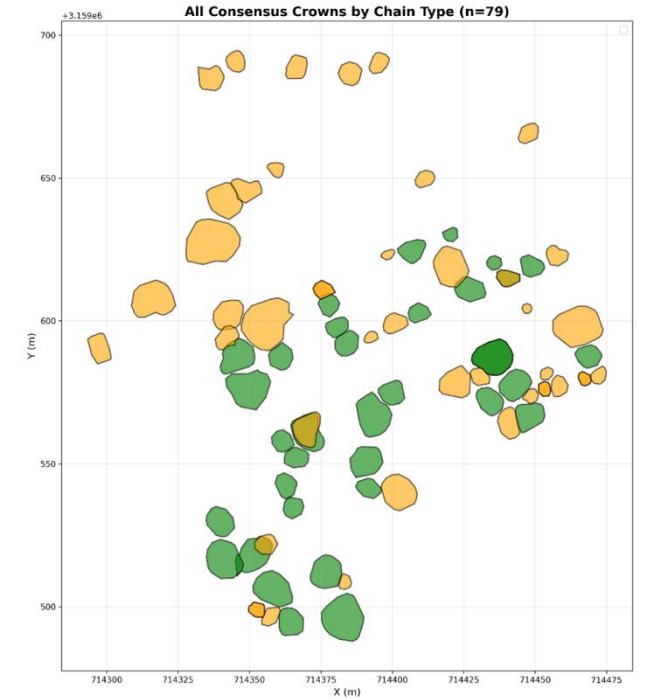
Consensus Crowns

- Individual crown detections vary across dates due to Phenological changes, illumination variations; Need a single polygon per tree for phenology analysis at fixed spatial locations
- Medoid: Select the crown polygon that minimizes total dissimilarity to all other detections in the chain
- Intersection Core: Compute overlapping region across all T detections; apply negative buffer to extract stable core
- We get same pixels sampled across dates and a single GeoDataFrame for GIS/satellite validation



Extracting Consensus Crowns From Smaller Chains

- Full-length chains provide ideal phenology time series, but cover only 35 trees. Can partial chains extend spatial coverage?
- No detection is bound to happen in high phenological changes, even in low confidence layers
- Construct consensus crown geometry using smaller chains, crop tree image from phase cross correlation aligned orthomosaics
- Quality Filtering: Chains must span ≥ 4 consecutive OMs, centroid std dev $< 8\text{m}$, avg pairwise similarity score < 0.55
- Adding length ≥ 4 chains $\rightarrow +44$ trees (+130% coverage)



Comparison with Ground Truth

GROUND TRUTH COMPARISON METRICS (IoU threshold = 0.3)

Precision: 0.645

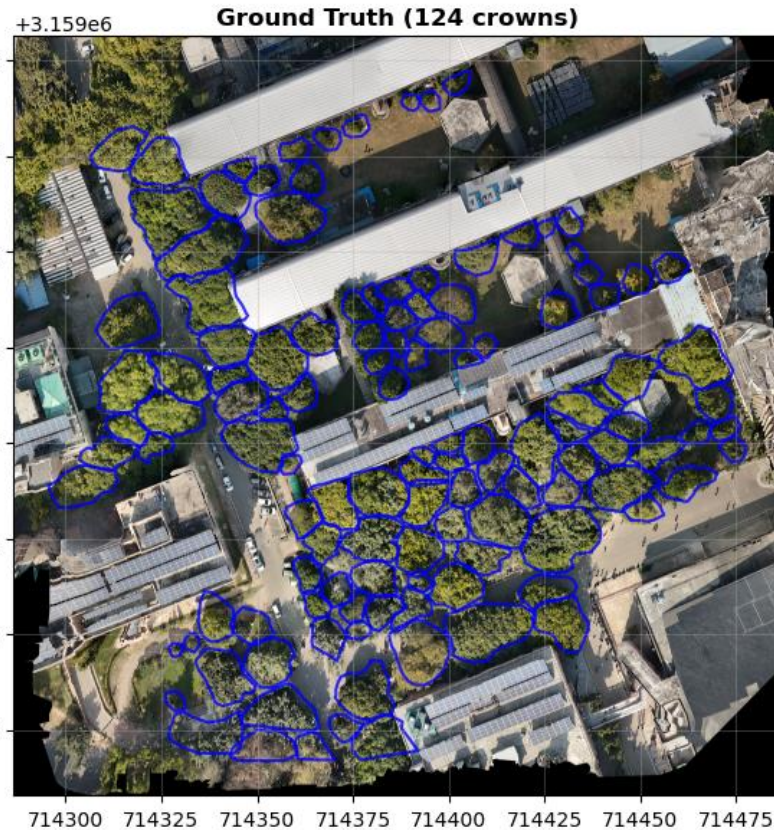
Recall: 0.161

F1 Score: 0.258

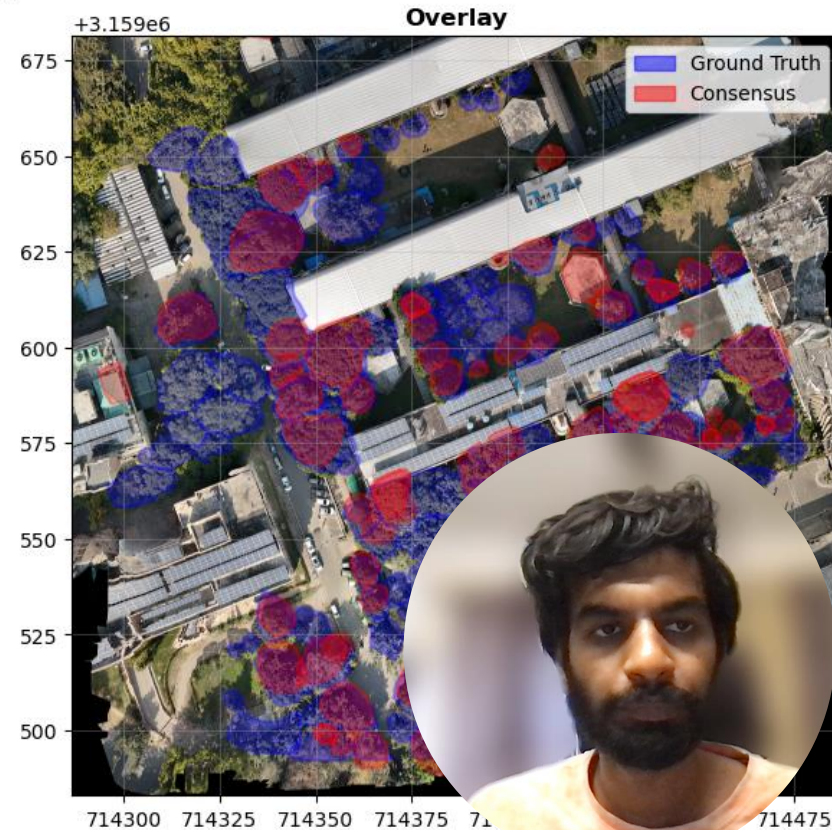
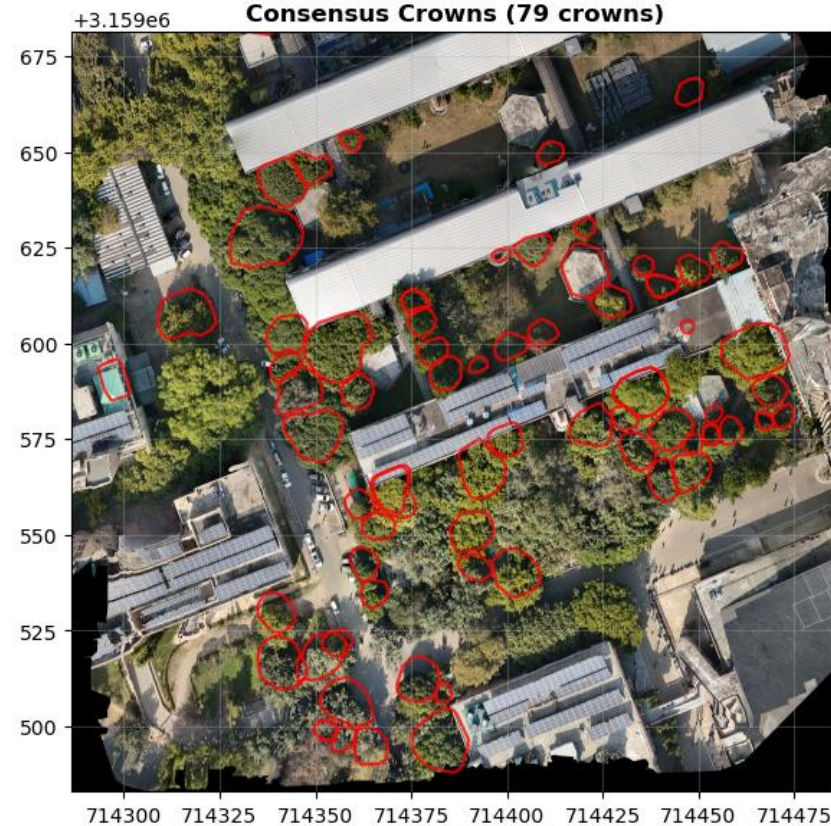
True Positives: 20

False Positives: 11

False Negatives: 104

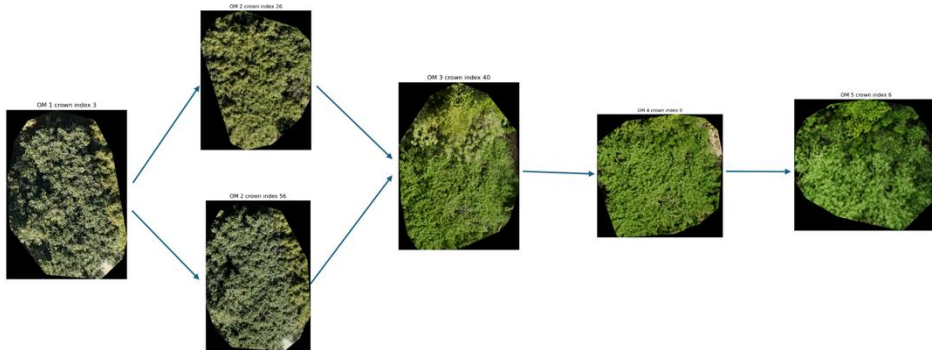


Ground Truth vs Consensus Crowns (79 tracked crowns)

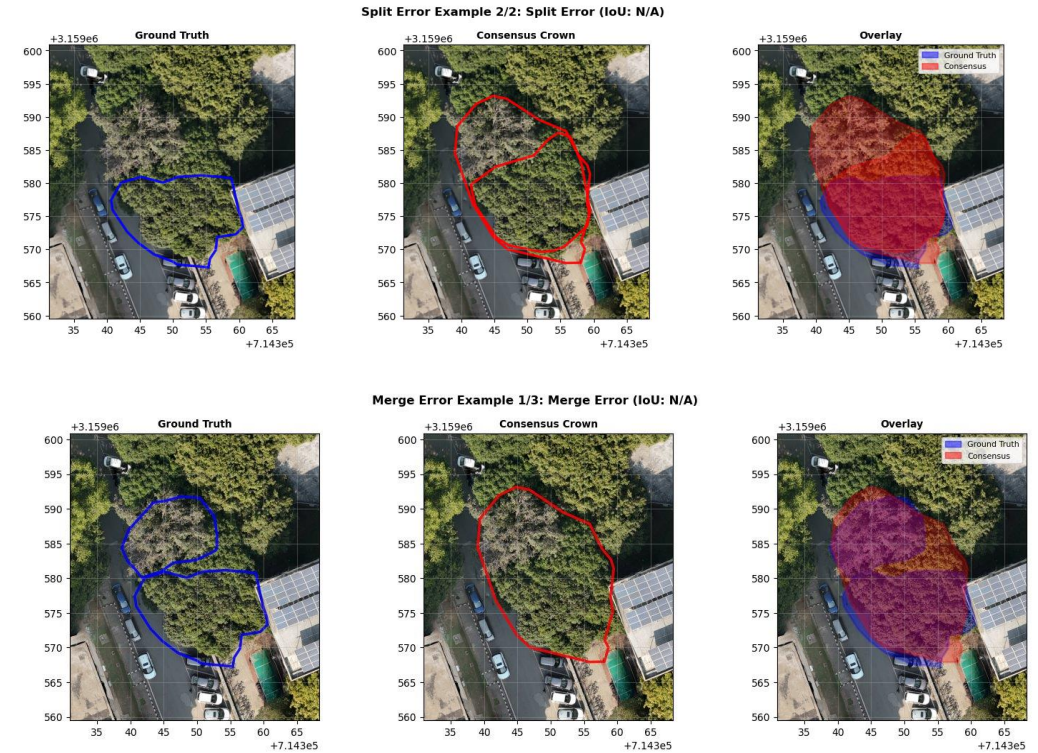
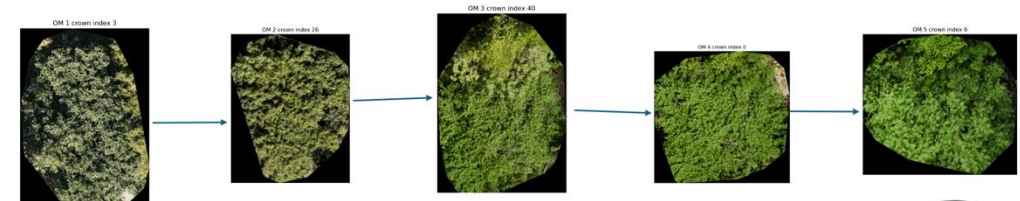


Split Merge Resolution

- For each candidate successor branch, look ahead W OMs and count whether at each OM the branches remain split (≥ 2 distinct nodes) or collapse to a single node.
- Compute `split_support` = number of OMs where branches remain split (or sum of branch counts) and `single_support` = number of OMs showing single continuity.
- Decide split wins when `split_support` > `single_support`



Resolved to



Phenological Metrics

- Vegetation signals from consensus crowns to quantify seasonal canopy dynamics of individual tree
- Chromatic Coordinates
 - GCC (Green Chromatic Coordinate): $G / (R + G + B) \rightarrow$ Greenness proxy
 - RCC (Red Chromatic Coordinate): $R / (R + G + B) \rightarrow$ Senescence indicator
- Vegetation Fraction (HSV Masking):
 - Hue: 65° - 173° (green range)
 - Saturation: ≥ 0.15 (color purity)
 - Value: ≥ 0.12 (exclude shadows)
 - Output: Ratio of vegetated pixels per crown



Article

Quantifying Leaf Phenology of Individual Trees and Species in a Tropical Forest Using Unmanned Aerial Vehicle (UAV) Images

John Y. Park ^{1,*}, Helene C. Muller-Landau ², Jeremy W. Lichstein ¹, Sami W. Rifai ^{3,4}, Jonathan P. Dandois ^{2,5} and Stephanie A. Bohlman ^{2,4}

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Article

Observing Spring and Fall Phenology in a Deciduous Forest with Aerial Drone Imagery

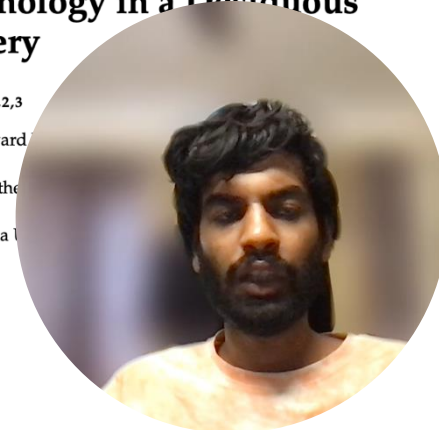
Stephen Klosterman ^{1,*} and Andrew D. Richardson ^{1,2,3}

¹ Department of Organismic and Evolutionary Biology, Harvard University, 73 Oxford Street, Cambridge, MA 02138, USA; Andrew.Richardson@nau.edu

² School of Informatics, Computing and Cyber Systems, Northern Arizona University, Flagstaff, AZ 86011, USA

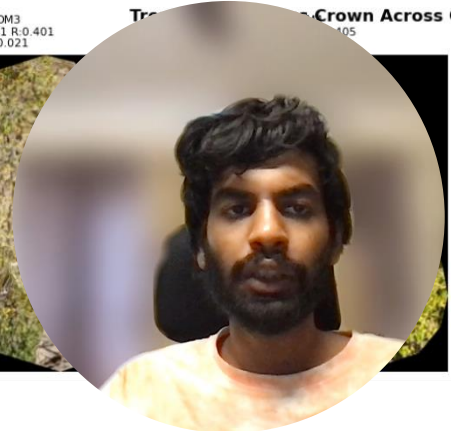
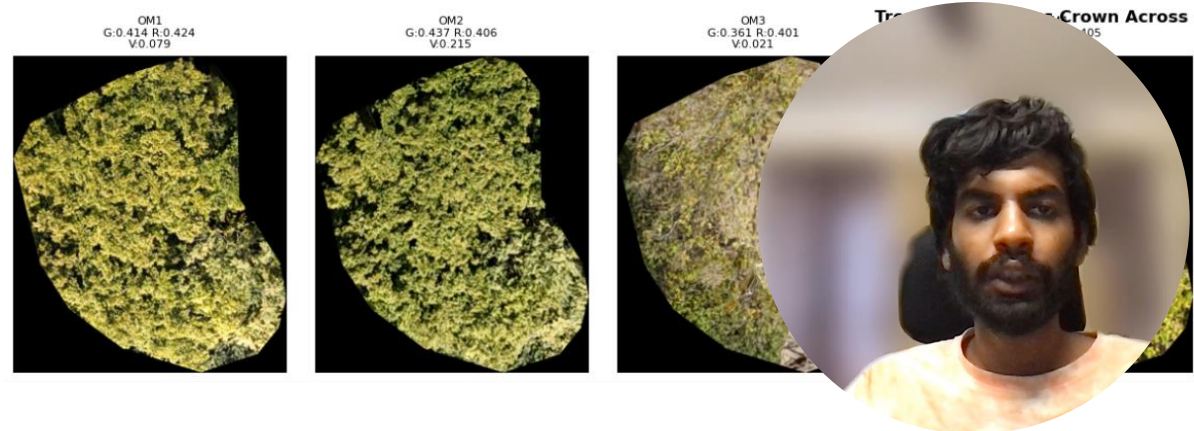
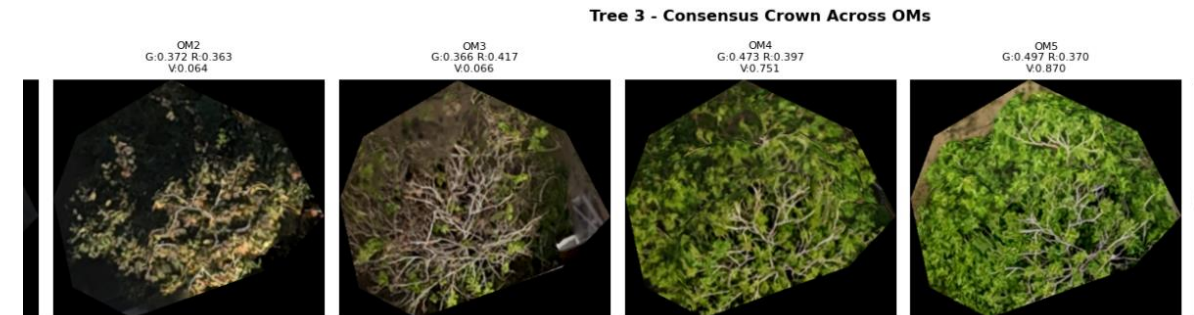
³ Center for Ecosystem Science and Society, Northern Arizona University, Flagstaff, AZ 86011, USA

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Phenological Metrics

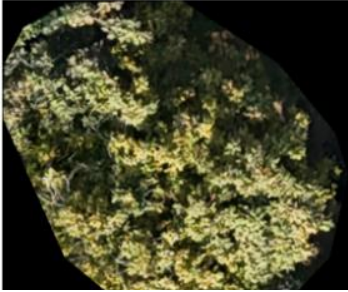
- Spatial Texture (Heterogeneity Proxies):
 - GLCM (Gray-Level Co-occurrence Matrix):
 - Contrast, Homogeneity, Energy (4 angles, $d=1$)
 - Gray Std Dev (smoothed): Intra-crown brightness variation
 - Shannon Entropy: Spatial randomness (texture complexity)
- Quality Control Flagging (Per Observation):
 - Valid pixel fraction < 60% (NoData/occlusion)
 - Shadow fraction > 55% (low illumination)
- Date-Wise Normalization: z-score
- Per-Tree Normalization: Min-max scaling



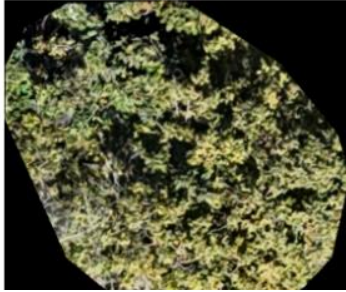
Phenological Patterns

Tree 14 - Consensus Crown Across OMs

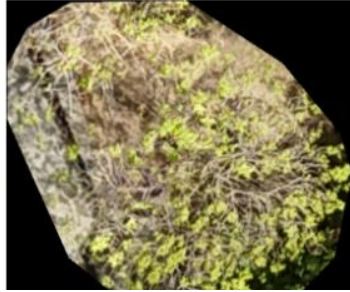
OM1
G:0.402 R:0.383
V:0.274



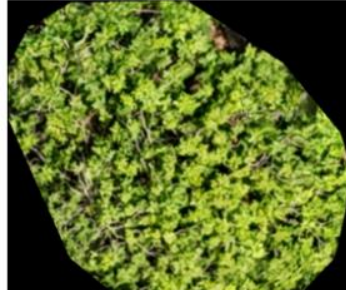
OM2
G:0.398 R:0.379
V:0.247



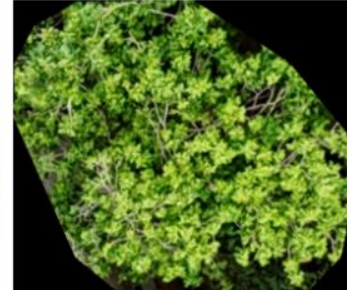
OM3
G:0.374 R:0.398
V:0.052



OM4
G:0.482 R:0.386
V:0.891



OM5
G:0.517 R:0.355
V:0.876



OM6
G:0.389 R:0.334
V:0.809

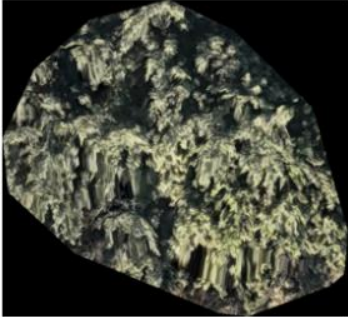


OM7
G:0.387 R:0.337
V:0.815

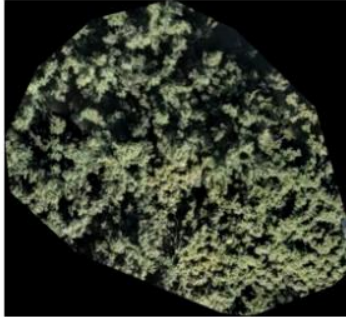


Tree 5 - Consensus Crown Across OMs

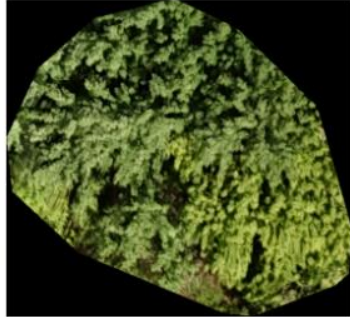
OM1
G:0.360 R:0.336
V:0.269



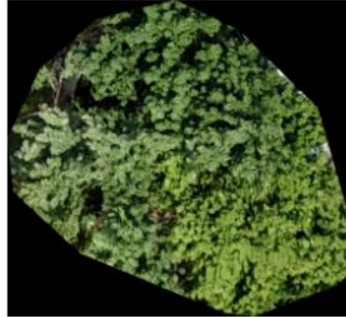
OM2
G:0.406 R:0.313
V:0.565



OM3
G:0.483 R:0.369
V:0.844



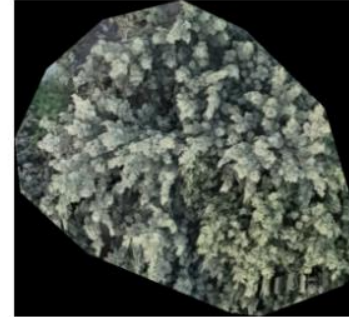
OM4
G:0.489 R:0.328
V:0.868



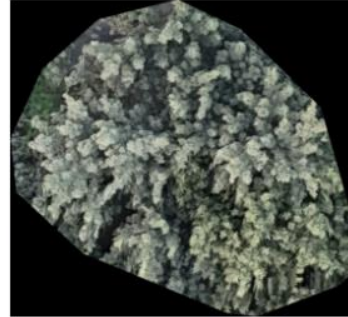
OM5
G:0.476 R:0.327
V:0.871



OM6
G:0.359 R:0.321
V:0.259



OM7
G:0.358 R:0.322
V:0.257



Tree 9 - Consensus Crown Across OMs

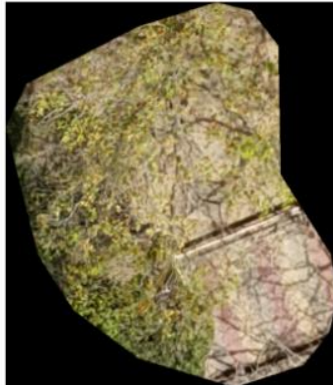
OM1
G:0.414 R:0.424
V:0.079



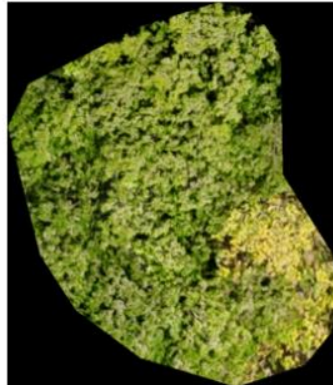
OM2
G:0.437 R:0.406
V:0.215



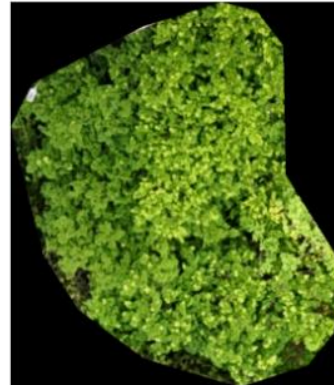
OM3
G:0.361 R:0.401
V:0.021



OM4
G:0.468 R:0.405
V:0.714



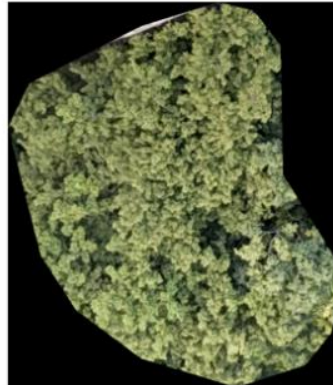
OM5
G:0.538 R:0.360
V:0.964



OM6
G:0.402 R:0.359
V:0.918

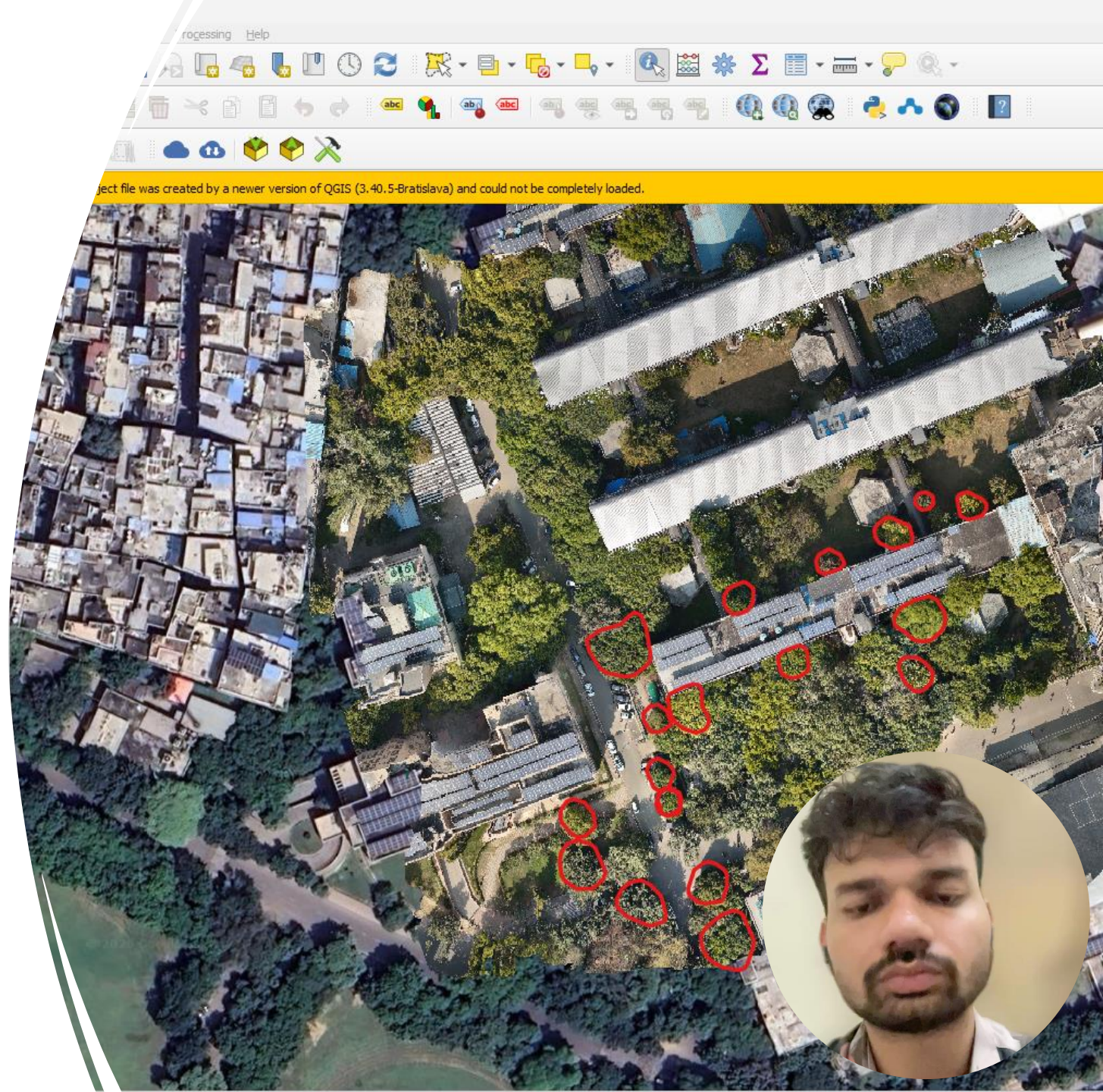


OM7
G:0.401 R:0.359
V:0.924



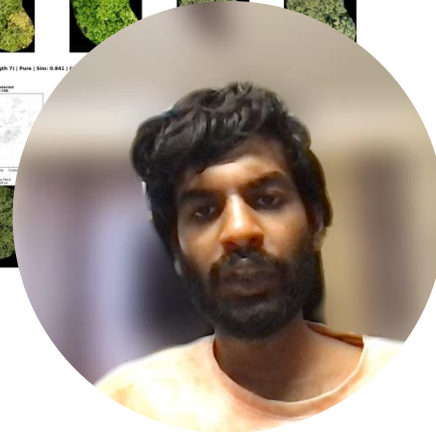
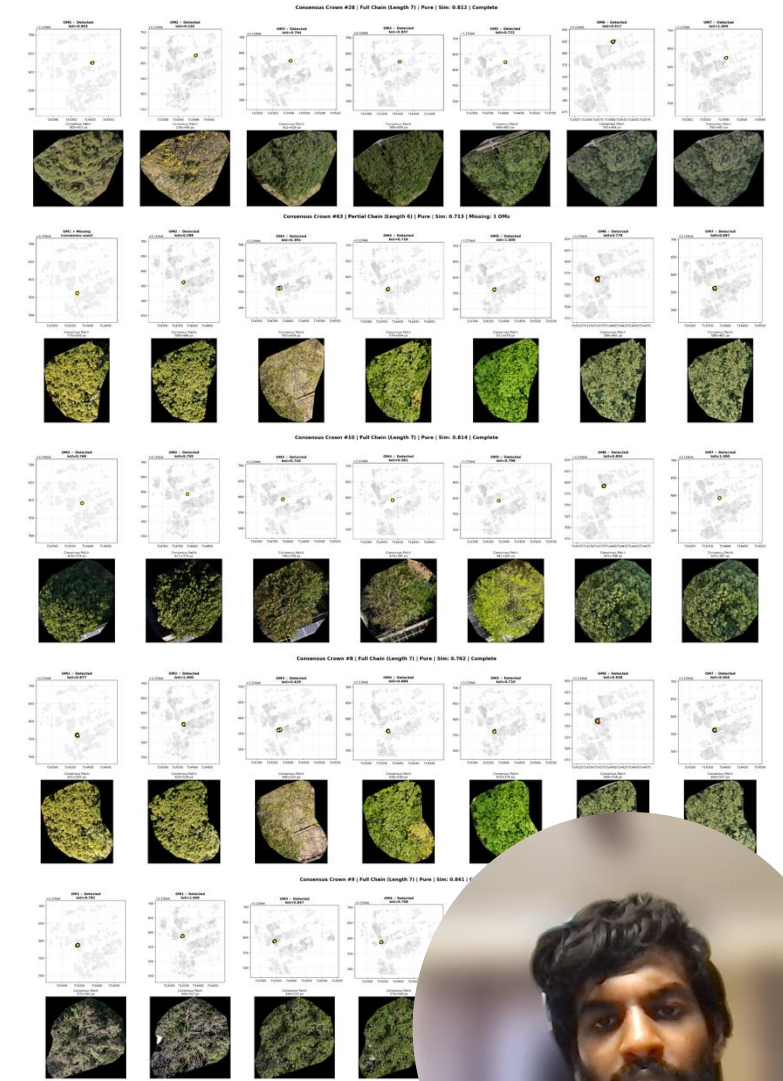
Qfield Ground Truth Marking

- The consensus crowns was uploaded to qfield app and then it was used to mark trees to find out which species is that particular tree and that will also be used to make our algorithm decide whether tree is leaf shedding , flowering or blooming



Future Work

- Post-consensus validation using CNN classifier on RGB patches to filter false positives (shadows, buildings, artifacts)
- Confidence slider: User-adjustable threshold controls tracking aggressiveness (precision vs recall trade-off)
- Classify each crown × date as "Leaf-On" or "Leaf-Off" using GCC, vegetation fraction, and texture features and generate monthly leaf cover vectors per tree
- Align these drone-derived leaf state labels with satellite data pixels towards satellite-based phenology models



Thank You

Questions?